

Introduction to Linux (1) — Exercises

Lab 1 — Basic commands `ls`, `cd`, `mkdir`, `rm`, ...

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Part 1 of 2

The basics of the terminal

Commands useful for the following exercises:

cd

cp

hostname

ls

less

mkdir

mv

pwd

rm

ssh

uname

wc

Commands are briefly explained in [linux_commands.pdf](#)

Exercise 0: Find out about commands

Find out *how to find out*:

Strategy: Look in `linux_commands.pdf`; the lecture notes (`linux_1.pfd`); search the net; ask somebody; just try something...

Last resort: There are solutions in `linux_1_solutions.pdf`

Also, try the following command:

`man <COMMAND>`



where you substitute `<COMMAND>` with the name of an actual command:

`man uname`

Scroll man-pages using the  and  keys or space.

Exit by  (*quit*)

Start the terminal

Open the terminal on the computer in front of you:

On a Mac, click the search icon (🔍) in the upper right corner, and type `terminal`

The terminal offers a prompt (command line), where you type all commands

Sidenote: On Windows, you can install WSL2 to run Linux on Windows.

Exercise 1 System info

It is assumed that you have a Linux-like terminal on the computer in front of you (such as a Mac). If not, skip this exercise.

1. What command is used to get the name of the computer you are logged into? Try it out
2. What command is used to get the name of the *operating system* (OS) of the computer? Try it
3. Add the *-a switch* (*flag, option*) to the above command. What do you see?

Exercise 2 Log in to remote server: `triton.dsv.su.se`

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1. What is the command for logging into another computer (remote server)?
 2. How do you log into `triton.dsv.su.se` from the terminal?
 3. Log into `triton.dsv.su.se`
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You should be able to use the same command from a Windows *PowerShell*.

Exercise 3 Repeat system info, different machine

After logging into `triton.dsv.su.se`:

1. Run the command `hostname`. What do you see?
2. Run the command `hostname -A` (uppercase A). What do you see?
3. Run the command `uname -a` (lowercase a). What do you see?

Exercise 4 The basics: list files

1. What is the command for listing the files in a directory? Run it
2. List all files in the `/bin` directory
3. List the contents of user `kali2818`'s home directory
4. List the contents of `texts/` in `kali2818`'s home directory

Exercise 5 The basics: move about

1. What is the command for printing the path to the current directory? Run it
2. What is the command for changing to another directory?
3. Move (change) to the `/tmp` directory
4. Run the command printing the current directory again (see 1.)
5. How do you move *back* from `/tmp` to your *home* directory?

Exercise 6 Create new directory

Run `cd`  to ensure that you are in your home directory.

1. What command creates a new directory?
2. In your home directory, make a new directory called `lab1`
3. List the contents of the current directory, to verify that `lab1` is now there (there should also be a directory called `public_html`)
4. List the contents of the new `lab1` directory (it should be empty)

Exercise 7 Copy a file (1/2)

1. What command is used for copying files?
2. Copy the file `ks1.txt` from the directory `/home/dsv/kali2818/texts/` (or shorter `~kali2818/texts/`) to your newly created `lab1` dir.
3. Verify that the file `ks1.txt` is now in `lab1`

Exercise 8 Copy a file (2/2)

1. Change directory to lab1/
2. Copy the file
~kali2818/texts/deeper_understanding.txt to your
current directory (lab1)

Hint: What character is used for “put it here” (to the current directory)?

Exercise 9 Rename a file

1. What command is used for moving and/or renaming files?
2. Rename the file `ks1.txt` (in directory `lab1/`) to `poem.txt`
(If you happen to be in a different directory than `lab1/`: first do `cd ~/lab1/`)
3. List files, to ensure that there is now a file `poem.txt`, but no `ks1.txt`

Exercise 10 `cp` vs. `mv`

1. What is the difference between `cp` and `mv`?
2. What is the difference between copying and moving a file?

Exercise 11 Viewing the contents of a file

1. What commands can you use to view the contents of a text file?
2. Take a look at the contents of `poem.txt` (previously `ks1.txt`, before you renamed it)
3. What's in the file `deeper_understanding.txt`?

Exercise 12 Removing files

1. What command is used for deleting files?
2. Return to your home directory
3. Remove all files with the `.txt` extension in the directory `lab1/`

Exercise 13 Copy a directory

1. Copy the directory `/home/dsv/kali2818/texts/` and its contents to `lab1/` (in your home directory)
2. List the contents of `lab1/` to ensure that the `texts/` directory is there
3. List the contents of the `texts/` directory

Exercise 14 Count lines

1. Is the `wc` command short for “water closet”?
2. Count the number of lines in the file `lab1/texts/ks1.txt`
3. Count the total number of lines in the files with a `.txt` extension in `lab1/texts/`

Hint: Consider using a joker character, to match several files in one go.

Exercise 15 man to find help

1. The `wc` command has a switch (*flag, option*) that makes it print a help message. Use the command

```
man wc
```

to find out what that switch is. You might have to scroll down to find it, use

 or . Use  to quit.

Run `wc` with the switch. (Notice that it is prefixed with *two* dashes, `--`.)

Many commands have this switch. Try it on some other commands, such as `uname` or `ls`.

Exercise 16 Scroll through output

Some commands generate more output than there may be room for in the terminal. One example is

```
python3 --help
```

which prints a long help message. (`python3` is used for running Python programs)

1. Run the command, to see what it looks like
2. Figure out a way to make it possible to scroll through the output, top to bottom

Hint: You need to somehow stick another command to the end of the command above. Look in `linux_commands.pdf` for clues.

Exercise 17 rm: file vs directory

1. Use `mkdir` to create a new directory:

```
mkdir testdir
```

2. Why doesn't the following command work for deleting `testdir/`?

```
rm testdir/
```

Exercise 18 Browsing a text file

1. What command can you use to browse a text file, e.g.
`/home/dsv/kali2818/texts/fk.txt`
2. How do you scroll down the text?
3. How do you exit?

Exercise 19 ...: The Only Way Is Up¹

(You can run the `cd` command, without arguments, to ensure that you are in your home directory.)

Given that you are in your home directory, what does the following command do?

```
ls ../
```

¹Yazz (1988)